Joshua Antu Kannappilly

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EDUCATION

University of Notre Dame — Notre Dame, IN
Bachelor of Science — May 2028
Major: Electrical Engineering | Concentration: Semiconductors & Nanotechnology — *3.89/4.00*

EXPERIENCE

IrishSat (CubeSat Design Team) – *Chief Electrical Engineer* — Jan 2025 – Present

- Leading a 20+ member team responsible for designing, prototyping, and testing the PCBs that power a 2U CubeSat for NASA's CubeSat Launch Initiative.
- Diagnosed a critical 1.8V vs. 3.3V I²C logic-level mismatch during bring-up of the prior-generation Battery Management System (BMS) board, identifying it as a fundamental architectural flaw.
- Led the redesign of the BMS in KiCad around the TI BQ40Z50-R2 to manage a 2S2P lithium-ion pack, resolving the mismatch and implementing SMBus communication with the central power management board for state-of-health monitoring.
- Wrote custom scripts to unlock BQ40Z50-R2 firmware, override default cell configurations, and programmatically trigger charge/discharge FETs, achieving the first successful demonstration of battery pack charging in club history.

NDnano – *Undergraduate Research Fellow* — May 2026 – July 2026

- Co-authored 2 paper submissions to the IEEE International Electron Devices Meeting (IEDM) as 3rd author, contributing experimental characterization and data analysis on FeRAM reliability.
- Executed full fabrication workflows in a cleanroom environment, gaining hands-on operation of advanced processing equipment to successfully build high-k MOSFET devices from scratch.

Bernstein Lab, University of Notre Dame – *Undergraduate Research Assistant* — Sep 2025 – Dec 2025

- Modernized a legacy Python library for the Newport SMC100 controller, implementing ESP-error handling and translating raw encoder counts into high-precision angular units (0.001° resolution).
- Integrated customized motion control drivers into the QCoDeS measurement framework, utilizing hardware abstraction to automate data logging into a centralized SQL database.
- Engineered a 2D "snaking" raster scan algorithm for Thorlabs Kinesis motors, optimizing spatial mapping efficiency and reducing mechanical wear for future LWIR thermal imaging.

MIDI Synthesizer – *Individual Project* — Jul 2025 – Aug 2025

- Engineered a MIDI Synthesizer, interfacing an ESP32 microcontroller with a USB Host Shield, PCM5102A DAC, and PAM8403D amplifier to drive dual 40mm speakers, enabling digital piano connectivity.
- Designed and routed a custom PCB in KiCad to integrate all components, optimizing for signal integrity and component layout for the compact synthesizer.
- Developed C++ firmware for the ESP32, implementing software-defined waveform generation for sine, square, triangle, and sawtooth waves, alongside real-time MIDI processing and polyphonic voice management.

LEADERSHIP

South Asian Student Association, University of Notre Dame – *Vice President* — August 2025 – Present

- Drove a 20-member increase in signups for Jazba, the club's signature performance showcase, by leading multiple community-building initiatives that deepened member engagement year-round.
- Generated 103 signups at the Activities Fair and represented South Asian culture at the MSPS Fair through cultural games such as carrom expanding the club's visibility across the broader Notre Dame community.

TECHNICAL SKILLS

Hardware: PCB Design, Arduino, ESP32, Raspberry Pi, Semiconductor Characterization, Semiconductor Fabrication
Programming: Python, C, C++, Cadence Virtuoso, LTspice, MATLAB, Verilog, RTOS, Automated Testing
Other: KiCad, Pandas, Matplotlib, Git, GitHub, Linux, SolidWorks, Rails, Automation Tools, Troubleshooting